

TODO WSTAW TYTUŁ PROJEKTU

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Opis zamierzeń badawczych ma zawierać

1. Wstępny tytuł rozprawy doktorskiej (do 300 znaków)
2. Wprowadzenie (określenie obszaru badawczego, powody wyboru problemu badawczego i jego znaczenie dla rozwoju danej dziedziny nauki)
3. Krótki przegląd literatury (kluczowe pozycje na temat danego problemu badawczego ze wskazaniem luk, które kandydat zamierza wypełnić, to nie jest bibliografia)
4. Główne pytania badawcze, które dezagregują na częściowe elementy stawiany przez kandydata problem badawczy
5. Metody badawcze do rozwiązania problemu badawczego
6. Oczekiwane teoretyczne i praktyczne wyniki problemu badawczego wybranego przez kandydata

1 Fisher-Pry jako gra różniczkowa - substytucja technologiczna

Fisher, J.C. and Pry, R.H. (1971) "A simple substitution model of technological change," *Technological forecasting and social change*, 3, pp. 75–88; Marchetti, C. (1977) "Primary energy substitution models: on the interaction between energy and society," *Technological forecasting and social change*, 10(4), pp. 345–356; Mercure, J.-F. (2012) "FTT: Power: A global model of the power sector with induced technological change and natural resource depletion," *Energy Policy*, 48, pp. 799–811

2 Differential Game of Resource Depletion and Energy Transition (?)

Hotelling, H. (1931) "The economics of exhaustible resources," *Journal of political Economy*, 39(2), pp. 137–175; Dasgupta, P.S. and Heal, G.M. (1979) *Economic theory and exhaustible resources*. Cambridge University Press; Karp, L. (2005) "Global warming and hyperbolic discounting," *Journal of public economics*, 89(2–3), pp. 261–282 – gra różniczkowa Long, N.V. (2011) "Dynamic games in the economics of natural resources: a survey," *Dynamic Games and Applications*, 1(1), pp. 115–148 – survey Benckroun, H., Breton, M. and Chaudhuri, A.R. (2019) "Mergers in nonrenewable resource oligopolies and environmental policies," *European Economic Review*, 111, pp. 35–52; Benckroun, H., Meijden, G. van der and Withagen, C. (2019) "An oligopoly-fringe non-renewable resource game in the presence of a renewable substitute," *Journal of Economic Dynamics and Control*, 105, pp. 1–20

3 Dynamic Bertrand Competition and Energy Transition (?)

Green, R.J. and Newbery, D.M. (1992) "Competition in the British electricity spot market," *Journal of political economy*, 100(5), pp. 929–953; Ludkovski, M. and Sircar, R. (2015) "Game theoretic models for energy production," *Commodities, Energy and Environmental Finance*. Springer

4 Metodologia

Başar, T. and Olsder, G.J. (1998) *Dynamic noncooperative game theory*. SIAM; Dockner, E. (2000) *Differential games in economics and management science*. Cambridge university press
Bressan, A. and Piccoli, B. (2007) *Introduction to the mathematical theory of control*. American institute of mathematical sciences Springfield

5 Ad 6) Oczekiwane wyniki

Oczekiwany jest, iż uzyskam warunków istnienia i charakterystyki równowag dynamicznych prowadzących do długookresowej substytucji technologii energetycznych. Analiza będzie obejmowała zarówno równowagi typu open-loop, jak i feedback, a także badanie ich stabilności oraz własności jakościowych rozwiązań.

Bibliography

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Ludkovski, M. and Sircar, R. (2015) “Game theoretic models for energy production,” *Commodities, Energy and Environmental Finance*. Springer.

Marchetti, C. (1977) “Primary energy substitution models: on the interaction between energy and society,” *Technological forecasting and social change*, 10(4), pp. 345–356.

Mercure, J.-F. (2012) “FTT: Power: A global model of the power sector with induced technological change and natural resource depletion,” *Energy Policy*, 48, pp. 799–811.